Mohammad Nasr

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Robotics PhD student building agentic AI architectures for conformance monitoring. Earlier, I worked on nonlinear dynamics and vibration developing a system for autonomous testing of vibrating systems that makes characterizing nonlinear, real-world dynamical behavior far more tractable. In acoustics, I have worked on binaural hearing, specifically on animals' auditory systems leading to potential bioinspired hearing designs. Available for a full-time internship, May–August 2027.

EDUCATION

Ph.D., Robotics & Autonomous Systems
Arizona State University · Expected May 2028
Fulton Fellow

M.Sc., Mechanical Engineering
Univ. of Nebraska–Lincoln · 2024
Advisor: Keegan J. Moore, Ph.D.
(Dynamics & Vibration)

B.Sc., Mechanical Engineering
University of Tabriz · 2021
Advisor: Mehran Mahboubkhah, Ph.D.

TECHNICAL TOOLKIT

ROBOTICS
ROS / ROS 2, Nav2, MoveIt, SLAM (ORB-SLAM3), EKF, Motion Planning, Manipulation, Sensor Fusion

SIMULATION
Isaac Sim, Gazebo, MATLAB/Simulink, Digital Twins

PROGRAMMING
Python, C++, MATLAB, Linux, Git

ML / AI
PyTorch, Vision-Language Models, LLMs, Reinforcement Learning

AGENTIC AI & VALIDATION
Multi-Agent Architectures, LLM Agents, Runtime Verification & Monitoring, Evaluation Pipelines, Model Context Protocol (MCP)

CONTROL / MATH
Optimization, System Identification, State Estimation

LINKS

mohammadnsr1.github.io
github.com/mohammadnsr1

RESEARCH EXPERIENCE

Agentic Verification for Robotic Systems — Current · PhD research
Arizona State University

- Designing a multi-agent runtime verification layer that independently checks robot behavior and intervene when actions drift from mission intent.
- Separated fast watchdog monitoring from slower foundation-model reasoning so verification never blocks real-time control; each agent boundary is justified by actor/checker independence, timescale separation or a certifiable authority limit.
- Built a working prototype in which agents act through tools exposed over the Model Context Protocol (MCP); first-author paper in preparation for IEEE Robotics and Automation Letters (RA-L).

Acoustic-Based, Bioinspired Robotic Platforms — Aug 2024 – Dec 2025
Arizona State University

- Analyzed hearing systems of domestic cats, arctic fox and aye-aye to design bio-inspired robotic hearing methods.
- Co-authored "Guided by Hearing," accepted in Materials Evaluation.

Autonomous Testing of Vibrating Systems — Aug 2022 – May 2024
M.Sc. research, University of Nebraska–Lincoln

- Developed a smart automatic modal hammer with a predictor–corrector excitation method, making experimental testing of nonlinear dynamical systems repeatable and autonomous.
- Published in the Journal of Vibration Engineering & Technologies (2024) and presented at IMAC XLII.

ROBOTICS PROJECTS

Active Perception for Object Localization — Jan – May 2026
Arizona State University · TurtleBot 4, ROS 2

- Built an active perception system pairing next-best-view selection with a confidence-scoring algorithm to improve pose-estimation reliability.
- Integrated Nav2 global/local planning with ORB-SLAM3, an EKF and a custom A* fallback for mapless navigation.

Assistive Follower Robot (Load Bearing) — Jan – May 2025
Arizona State University · TurtleBot 4, ROS 2

- Built a follower robot with a depth camera and LiDAR for hands-free tracking and obstacle avoidance.
- Trained a lightweight ResNet18 model for real-time shoe detection on a Raspberry Pi.

Solving a 4×4 Maze with MyCobot Pro 600 — Aug – Dec 2024
Arizona State University

- Implemented A* for maze solving and robot motion planning, verified in a digital-twin simulation.

MORE PROJECTS

RL for Robot Navigation (Q-Learning)
Univ. of Nebraska–Lincoln · Jan–May 2024

Autonomous Navigation, F1TENTH Platform
Univ. of Nebraska–Lincoln · Jan–May 2024

Robotic Arm Soccer Player
Univ. of Nebraska–Lincoln · Aug–Dec 2023

Planar Circular Restricted Three-Body Problem
Univ. of Nebraska–Lincoln · Jan–May 2023

AVIATION TRAINING

B1, Airframe Mechanics & Aircraft Maintenance Technology
Civil Aviation Technology College (CATC), Tehran · Sep 2020–Jul 2021
Passed civil aircraft maintenance coursework with EASA modules covering aerodynamics, structures, gas turbine engines and propeller systems.

LEADERSHIP

Local Committee Vice President
AIESEC, Tabriz · May–Sep 2017
Built volunteering projects that brought international students to Iran; negotiated financial and accommodation support with local partners.

JOURNAL PUBLICATIONS

Nasr, M., Masurkar, N., Meshkizadeh, P., & Dehghan-Niri, E. (2025). Guided by Hearing: Robotic Motion Planning for Precision in Gas Leakage Localization. *Materials Evaluation*. Accepted for publication.

Nasr, M., Singh, A., & Moore, K.J. (2024). Smart Automatic Modal Hammer: Predictor–Corrector Approach for Accurate Excitation of Dynamical Systems. *J. Vib. Eng. Technol.* 12 (Suppl 1), 1045–1063. doi.org/10.1007/s42417-024-01461-6

In preparation: Nasr, M. Watchdog: Physically Grounded Conformance Monitoring for Agentic Robotic Systems. For submission to *IEEE Robotics and Automation Letters (RA-L)*.

CONFERENCE PRESENTATIONS

Nasr, M., Masurkar, N., Meshkizadeh, P., & Dehghan-Niri, E. "Gas Leakage Localization Through Bioinspired Motion Strategy on Robotic Platforms," ASNT Research Symposium 2025, Indiana, June 23–27.

Nasr, M., Singh, A., & Moore, K.J. "Smart Automatic Modal Hammer for Studying Nonlinear Dynamical Systems," International Modal Analysis Conference XLII, Orlando, FL, Jan 29–Feb 1, 2024.

TEACHING EXPERIENCE

Teaching Assistant, Robotic Systems 1 Jan 2026 – Present
Dept. of Manufacturing & Systems, Arizona State University
- Conduct lab sessions using Dobot robotic arms; designed lab exercises and final project assignments.
- Grade student homework assignments and lab reports.

Teaching Assistant, Engineering Dynamics Aug 2023 – May 2024
Mechanical & Materials Engineering, Univ. of Nebraska–Lincoln
- Conducted recitation sessions and coached students on course content and problem-solving procedures.
- Corrected and graded student homework problems and exams.

TA, Kinematics & Dynamics of Machinery; Machine Design Aug 2022 – May 2023
Mechanical & Materials Engineering, Univ. of Nebraska–Lincoln
- Addressed student questions on course materials and solutions; corrected and graded homework and exams.

PROFESSIONAL EXPERIENCE

Aerial Robotics Team Leader & Design Lead Dec 2017 – Jun 2019
Robotic Lab, University of Tabriz, Iran
- Led the aerial robotics team for the Iran Open Robotics competition: designed the quadrotor frame, flew as pilot, and secured financial and lab support from research center supervisors.
- Designed fuselage, wings and tail in SolidWorks, analyzed aerodynamics in ANSYS, and built the first balsa-wood prototype for the national model airplane competition.
- Optimized an airfoil into a tapered wing, later built by 3D printing and balsa wood for the wind tunnel teaching lab.

Cofounder & Chief Product Designer May – Sep 2021
VISION Company, Tabriz, Iran
- Designed automatic hand and foot disinfectant devices in SolidWorks, adopted in schools, restaurants and stores; programmed an Arduino Uno as the main control unit.
- Designed an automatic face-mask delivery machine for malls, bus stops and train stations; product sales generated nearly 30% revenue.